

Clinical Risk Assessment Report: Patient 70

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.45x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.4%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.27 [Avg (34th %ile)] (-2.4%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.06 [Top 22%] (-1.2%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.54 [Avg (30th %ile)] (-1.0%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Patient Age (Age) **Value: 62.0 [Avg (43th %ile)] (-0.8%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.44 [Avg (66th %ile)] (-0.6%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.0 [Top 20%] (-0.5%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -1.58 [Bottom 9%] (-0.5%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: -1.55 [Bottom 10%] (-0.5%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Machine Learning Feature Attribution (SHAP Decision Path)

